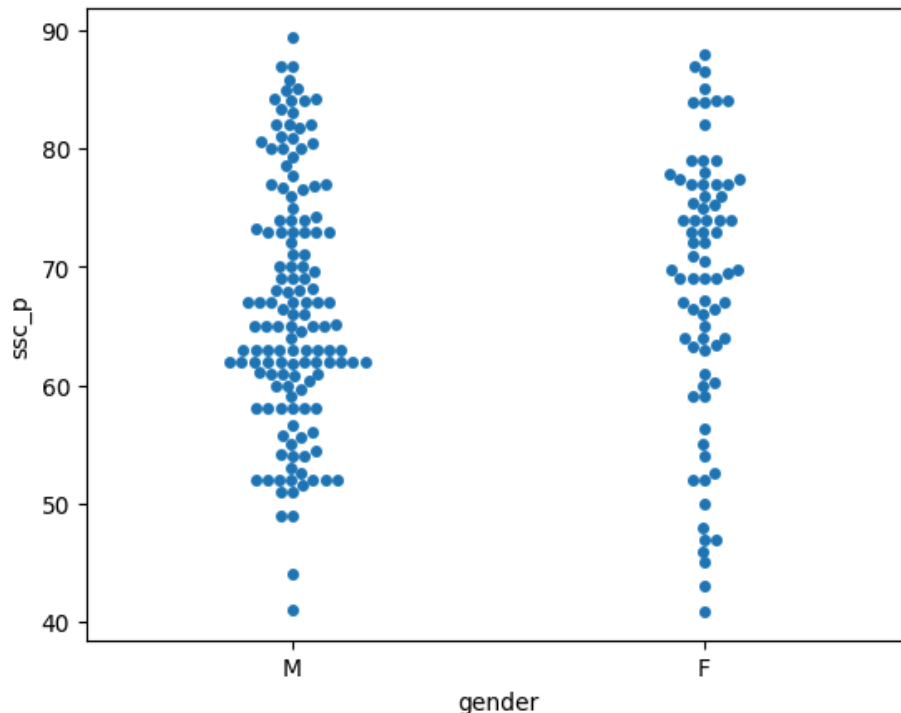


Seaborn Summary

Swarm_Plot:



-
- To visualise the distribution of numeric variables across different categories while showing every single data point. [\[1, 2, 3\]](#)
- It is essentially a categorical scatter plot where the data points are automatically adjusted along the categorical axis so that they **do not overlap**

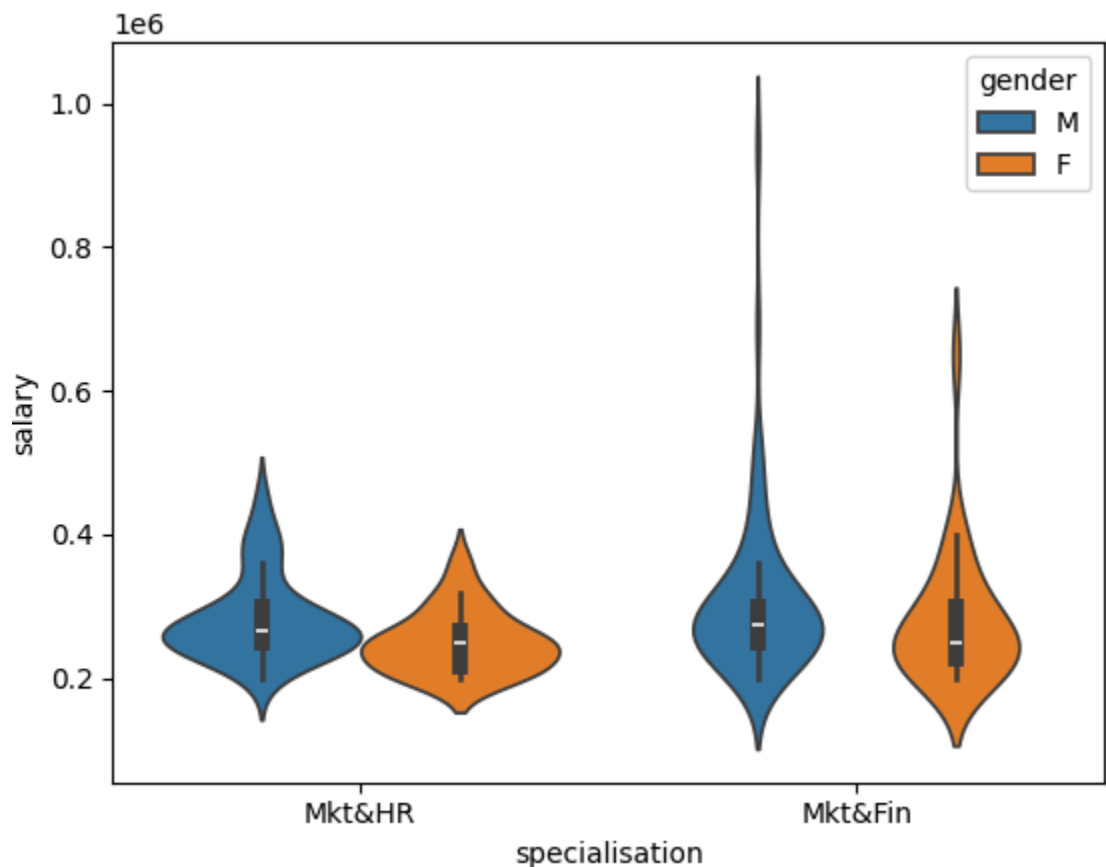
Here is exactly when and why you should use a swarm plot:

- 1. Primary Uses:
 - ❖ Showing Data Density - spreading the points out horizontally to form a "swarm" shape.
 - ❖ Comparing Categories - comparing test scores across different classrooms
 - ❖ Visualising Small to Medium Datasets - to see the literal footprint of your entire dataset.

2. Swarm Plot vs. Other Plots:

- ❖ Swarm Plot vs. Strip Plot (jitter=True): A strip plot uses random noise to spread points out, meaning some dots might still overlap by accident. A swarm plot uses a deterministic algorithm to ensure absolutely zero overlapping.
- ❖
- ❖ Swarm Plot vs. Box Plot: A box plot only shows summary statistics (like the median and quartiles) and hides the individual shapes of your data. A swarm plot shows you every single real value.
- ❖
- ❖ Swarm Plot vs. Violin Plot: A violin plot uses a mathematical curve to estimate density, which can look misleading with very little data. A swarm plot shows the true density using the actual data points

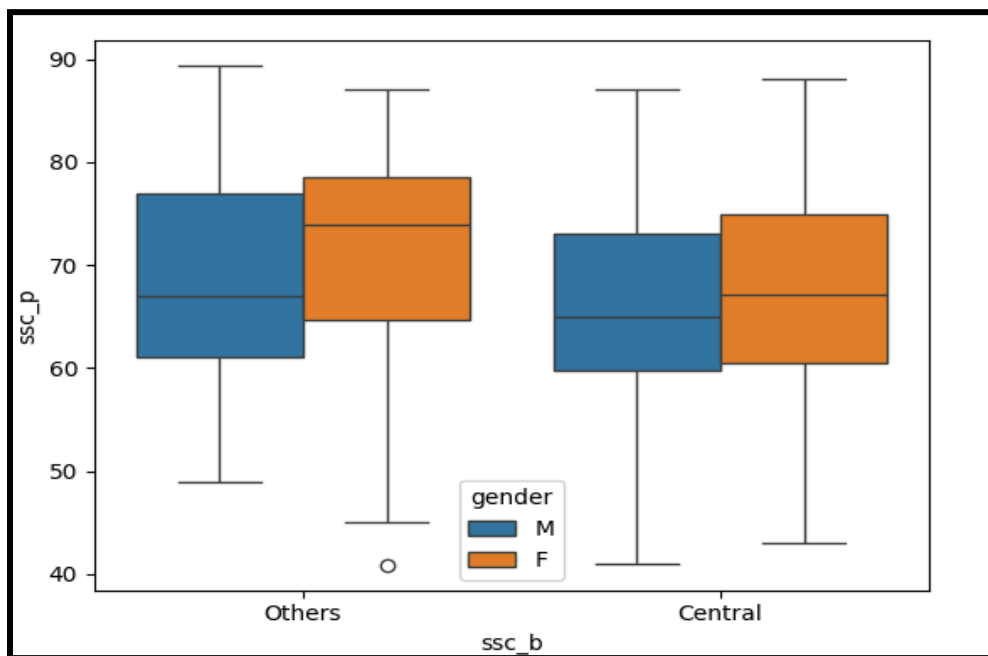
Violin Plot



here violin plot shows density of salaries they paid depending upon specialization MKd&Fin and Mkd&hr students (male and female)

1. Mkd&fin Male students got high salary nearly(10,00000) than others in specialisation(Mkd&Fin -female and Mkd&Hr - male,female)
2. Second most paid students are in MKd & Fin -female (790000)
3. In both specialisation Mkd&Fin(Male&female) got low minimum salary as 125000 and Mkd&Hr (nearly more or less equal 1,75000)
4. Mkd & Hr got (Male -590000) higher than females(400000) in the same specialisation.
5. High density of male students in Mkd&Hr paid salary from(2,20000 to 3,70000) for females in same group got nearly same
6. For Mkd&Fin high density of male students and females paid nearly the same salary from(2,20000 to 3,70000) .

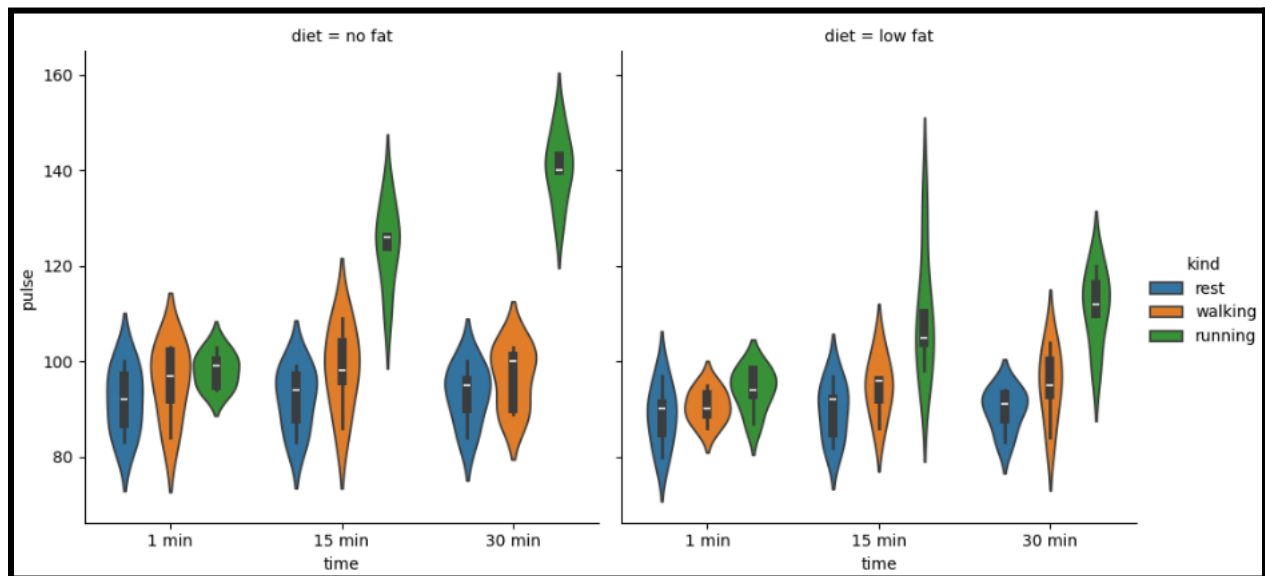
Box Plot:



- ❖ The box plot always denotes the percentile of that.
- ❖ The males in the Other of ssc_b scored higher marks as Q4-89 than the others .

- ❖ The females in both Others and Central of ssc_b scored the same marks Q4 - as 87.
- ❖ The initial scoring marks of Others in the males was Q1- 49 and females in the same board got marks as Q1 - 45.
- ❖ Females in the others got median marks Q3 as 74 which is higher than the others male and central students
- ❖ The females in the board of others had lesser outliers as 40.
- ❖ The central board males got less initial marks as 42.
- ❖ The males in the Other of ssc_b got Q2 - 60-67; Q3 - 67-77; IQR of others in the male is 60 - 77;
- ❖ The female in the Other of ssc_b got Q2 - 65 - 75; Q3 - 75 - 80; IQR of others in the female is 65 - 80;
- ❖ The central board male got median marks as Q3- 65 IQR -58 - 74; and for females got median as Q3 - 68;IQR- 62 - 75;

Catplot:



Here the cat plot shows violin plot and Box plot

Here the violin plot shows the density of the kind of person depending on their diet with and without fat consumption and doing some exercise with various time intervals.

Diet=No fat:

Rest Person:

-
- The blue one denotes rest taking people with no fat diet having a median heart pulse rate of 92.
- Here high density pulse occur between 85-100 within 1 min with rest person and little bit variation each 15 min
- After 15 mins the person who got a median pulse of 95 pulse rate
- After 30 mins the pulse of that person was nearly the same in the previous 15 mins.
- But for each interval of period that person's high pulse rate is 110 range; minimum pulse is 70 .

Walking Person:

-
- The person who is walking with no fat diet takes their min pulse is 70 ;max pulse is 118 at 1 min of exercise.
- Here high density pulse occur between 88-95 within 1 min with walking person 95-98 for 15min of exercise and for 30 mins of exercise 95 to 100;
- After 15 min of exercise their max pulse rate changes to 118 where min pulse is 110.
- After 30 min of exercise their median pulse rate is 118; max pulse rate changes to 120 where there is change to 78as min pulse.

Running Person:

- But for a running person who got a minimum pulse as 92 ;max as 110 at 1 min of exercise;
- Here high density pulses occur between 98-100 within 1 min with a running person after 15 min and 30 min of exercise is pulsed at 105-110; 138 -143 pulse.
- After 15 min of exercise median pulse rate change to 128;maximum pulse rate is 155; at min pulse is 100
- After 30 mins of exercise compared to others its max pulse rate is as high as 160;min as120; median pulse rate is 140.

Diet=low fat:

Rest Person:

-
- The blue one denotes rest taking people with a low fat diet having a median heart pulse rate of 92.
- Here high density pulse occur between 85-100 within 1 min with rest person and little bit variation in 15 min ;after 30 min it becomes 95-98;
- After 15 mins the person who got a median pulse of 95 pulse rate .
- But for each interval of period that person's high pulse rate is 96 range; minimum pulse is 82 .comparing to no fat diet rest person their min pulse is high max pulse low to this low fat diet person

Walking Person:

-
- The person who is walking with no fat diet takes their min pulse is 85 ;max pulse is 100 at 1 min of exercise.
- Here high density pulse occur between 95-104 within 1 min with walking person 98-105 for 15min of exercise and for 30 mins of exercise 95 to 100;
- After 15 min of exercise their max pulse rate changes to 125 where there is no change in min pulse.
- After 30 min of exercise their median pulse rate is 100; max pulse rate changes to 115 where there is change to 80as min pulse.

Running Person:

- But for a running person who got a minimum pulse as 80 ;max as 108 at 1 min of exercise; for 15 mins and 30 mins of exercise min & max respectively 80-150; 98-139;
- Here high density pulse occur between 95-100 within 1 min with running person after 15 min and 30 min of exercise is pulsed at 125-128; 108 -120 pulse
- After 15 min of exercise median pulse rate change to 128;maximum pulse rate is for 30 mins exercise 110; at1 min pulse is 95
- After 30 mins of exercise max pulse rate is 130;min as 90;

Conclusion:

- Finally Wit and Without fat diet taken rest person without doing any exercise nearly got their min and max pulse rate same
- As for walking, a person's max pulse is 120 major variations. In a low fat person after 1 min exercise their max pulse is only 98 and there is little difference there.

- The vital changes is happend running persons heart pulse each min of exercise with no fat diet their minimum maximum pulse has varied here. Their max pulse is 160 compared to other this no fat 30 min running person got High max pulse minimum heart pulse is 85.